

GEN AI :

Perfect! Let's make a **structured interview prep set** for your tech stack: **LangChain4j, AI models, RAG, Weaviate DB, and Spring Boot**. I'll break it into **Beginner → Medium → Advanced** levels.

Beginner Questions

LangChain4j & AI Models

1. What is LangChain4j, and how does it help in AI applications?
 2. What is a language model, and how is it used in applications?
 3. What is RAG (Retrieval-Augmented Generation)?
 4. What are embeddings, and why are they important for semantic search?
 5. What is Weaviate DB, and what type of database is it?
 6. How does Weaviate store and query vector embeddings?
 7. What is a Spring Boot application, and why do we use it with microservices?
 8. What is a Kafka topic, and why is it used in AI pipelines?
 9. Explain the difference between structured and unstructured data in AI applications.
 10. How do you send data from Spring Boot to Kafka?
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Medium-Level Questions

LangChain4j & RAG

1. How does LangChain4j orchestrate multiple AI services?
2. What is the difference between a **retriever** and a **generator** in a RAG pipeline?
3. How do you integrate vector DB (like Weaviate) with LangChain4j for document retrieval?

4. Explain the role of embeddings in a RAG system.
 5. How do you handle multiple documents in RAG using LangChain4j?
 6. What is prompt engineering, and how is it applied in LangChain4j pipelines?
 7. How do you deploy a Spring Boot RAG service to Docker?
 8. How do you handle scaling multiple AI services in a Spring Boot microservices architecture?
 9. How do you monitor AI model latency and performance in production?
 10. How do you ensure message reliability when sending queries via Kafka to AI services?
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Weaviate DB

1. How do you create a schema in Weaviate for storing documents?
 2. How do you perform semantic search using Weaviate?
 3. How does Weaviate handle vector similarity search?
 4. How do you integrate Spring Boot with Weaviate using REST API?
 5. How do you update or delete embeddings in Weaviate?
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Advanced-Level Questions

LangChain4j & RAG

1. How do you optimize a RAG pipeline for high-throughput query processing?
2. How do you handle multi-turn conversations using LangChain4j?
3. How do you combine multiple retrievers for different document sources?
4. How do you ensure AI model responses are contextually relevant in a RAG system?
5. How do you handle embeddings versioning and updating in production?
6. Explain latency vs accuracy trade-offs in a RAG system.
7. How do you implement caching of retrieved documents for efficiency?

Weaviate & AI Integration

1. How does Weaviate scale horizontally for large vector datasets?

2. How do you secure Weaviate API endpoints for sensitive data?
3. How do you handle real-time ingestion and indexing of new documents in Weaviate?
4. How do you measure similarity search performance and accuracy in Weaviate?
5. Explain the process of integrating multiple vector DBs in a single AI pipeline.

Spring Boot & Microservices

1. How do you design microservices for AI pipelines (API, RAG, Summarizer)?
2. How do you handle distributed transactions and message ordering with Kafka?
3. How do you monitor and log AI services running in Spring Boot?
4. How do you handle retry, fallback, and error handling in AI microservices?
5. How do you deploy and orchestrate multiple AI services using Docker Compose?

Pro Tip for Interviews:

- Focus on **RAG architecture**: Retriever → Vector DB → Generator → Summarizer.
- Know how **Kafka, Spring Boot, Docker, Weaviate, and LangChain4j** connect together.
- Be ready to discuss **scaling, monitoring, and latency trade-offs** in AI microservices.

If you want, I can **next create a full set of answers for beginner and medium questions** like I did for Docker and Kafka, so you have **ready-to-go material for interviews**.

Do you want me to do that?

Beginner Questions:

 Medium-Level Questions:

 Advanced-Level Questions